



SQL DATABASE DEVELOPMENT KNOWLEDGE

Hint

If you don't have PostgreSQL installed or want to quickly test your SQL, use the Programiz Online SQL Compiler:

Programiz SQL Online Compiler

Phase 1 Watch Videos

Students must watch the following videos before starting the practical tasks.

Video to Watch

DATABASE - PostgreSQL Tutorial for Beginners

<https://www.youtube.com/watch?v=WtA1pfOz5uU>

INSTALLATION:

https://youtu.be/UL05e_8JIQA?si=4RsdUSIZ0Q574CnM

SQL COMPLETE COURSE

<https://youtu.be/hlGoQC332VM?si=rFGLdSTvte0UmFfP>

CREATE YOUR FIRST DATABASE:

<https://www.youtube.com/watch?v=T62SzpkipoE>

Databases for AI Automation: Connect n8n with Supabase

<https://youtu.be/k-QDAjmFSYo>

How to use Supabase PostgreSQL in n8n

<https://youtu.be/OWQ98EZdns0?si=bvlyPKxeWMXKiAMf>

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Phase 2 Install & Setup

Students should:

1. Install **PostgreSQL**.
2. Install **pgAdmin**.
3. Create a PostgreSQL server connection.
4. Understand the following:

- Server
- Database
- Schema
- Table
- Row
- Column
- Data Type
- Primary Key
- Foreign Key
- NULL
- Relationships
- CRUD
- SQL Query

Phase 3 SQL Practical Tasks

Task 1 Build a Lead Management Database

Create a table called:

leads

The table should contain at least:

id
name
email
phone
company
source
lead_score
status
created_at

Requirements

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Write SQL queries to:

1. Create the table.
2. Insert **at least 10 leads**.
3. Display all leads.
4. Display only qualified leads.
5. Display leads having a score greater than 70.
6. Update one lead's status.
7. Delete one lead.
8. Display the top 5 leads according to lead_score.

Do not simply copy a solution from the internet. Write the query yourself, run it, and understand the output.

Task 2 SQL Sales Analytics

Using your leads table from Task 1, write *SQL queries* to answer:

Question 1

How many total leads are there?

Question 2

How many leads are:

New
Contacted
Qualified
Converted
Lost

Question 3

What is the average lead score?

Question 4

Which lead has the highest score?

Question 5

How many leads came from each source?

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For example:

Facebook	12
LinkedIn	8
Website	15
Google	7
Referral	5

Question 6

Display all leads with a score between **50 and 80**, sorted from highest to lowest.

Concepts students should demonstrate

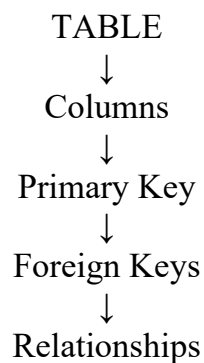
SELECT
WHERE
ORDER BY
LIMIT
COUNT()
AVG()
GROUP BY
BETWEEN

Phase 4 Database Design on Paper

For these tasks, *do not write SQL*.

Students must design the database *on paper* first.

They should draw:



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Task 3 AI Automation CRM

Scenario

You are building an **AI-powered CRM** for a digital marketing agency.

The system receives leads from:

- Website
- Facebook
- Instagram
- LinkedIn
- WhatsApp

An AI agent analyzes every lead and assigns:

- Lead score
- Lead status
- Lead category

A customer may have multiple conversations with the AI agent.

Each conversation may contain many messages.

Your task

Design the database on paper.

You should determine the required tables.

At minimum, your design should handle:

Customers

Leads

Conversations

Messages

For every table, identify:

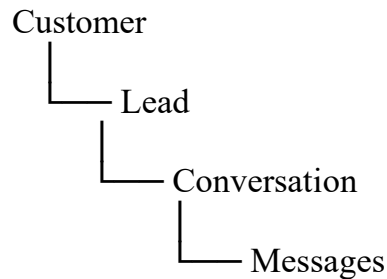
- Table name
- Columns
- Data types
- Primary Key
- Foreign Key

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Then draw the relationships.

For example:



Task 4 AI Customer Support Knowledge System

Scenario

A company wants to build an **AI Customer Support Agent**.

Customers can ask questions through a chatbot.

The system needs to store:

- Customer information
- Support tickets
- Conversations
- Messages
- Products
- Orders
- FAQs

The AI agent should be able to look at a customer's previous conversations and order information before answering.

Your task

Design the complete database structure **on paper**.

Determine:

1. What tables are required?
2. What columns does each table contain?
3. What is the Primary Key?
4. What are the Foreign Keys?

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5. Which tables have one-to-many relationships?
6. Which tables are independent?
7. Which tables should be connected?

Your final paper should contain an **ER-style database diagram**.

Bonus

Design the structure so that an AI agent can answer:

"What was my previous order and why was my support ticket opened?"

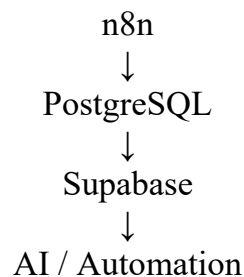
Phase 5 n8n + PostgreSQL + Supabase

Now students move from **database development** into actual **AI Automation development**.

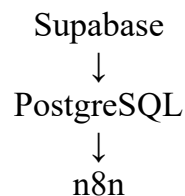
Important requirement

For every task, students must use **both PostgreSQL and Supabase**.

They must understand how data moves between:



and, where appropriate:



Task 5 Lead Capture Automation

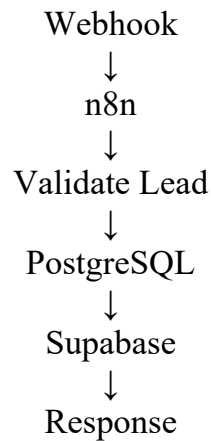
Scenario

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A website sends a new lead to n8n through a webhook.

Build this workflow:



Requirements

When the webhook receives:

name
email
phone
company
message
source

the workflow must:

1. Receive the lead through a Webhook.
2. Validate the required fields.
3. Insert the lead into PostgreSQL.
4. Store/synchronize the lead in Supabase.
5. Return a successful response.
6. Handle invalid/missing data.

Deliverables

Students must submit:

- n8n workflow screenshot
- PostgreSQL table screenshot
- Supabase table screenshot
- Sample execution

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- SQL used

Task 6 AI Lead Qualification + Database

Scenario

A lead enters your system.

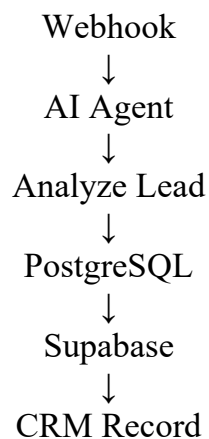
An AI agent must analyze the lead and determine:

lead_score
lead_status
lead_category

Example:

Lead Score: 87
Status: Qualified
Category: High Intent

Workflow



Requirements

The workflow must:

1. Receive a lead.
2. Send the lead information to an AI model.
3. Generate a score from 0–100.
4. Generate a status.
5. Generate a category.
6. Save the original lead.

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7. Save the AI analysis.
8. Store the final record in PostgreSQL and Supabase.

Bonus

Add an IF node:

Score $\geq$ 70



Send qualified lead notification

Task 7 AI Customer Conversation Memory

Scenario

Build a customer support AI agent that remembers previous conversations.

The system should store:

customer
conversation
messages

Workflow

Chat/Webhook



Find Customer



PostgreSQL / Supabase



Retrieve Previous Messages



AI Agent



Generate Response



Save New Message

Requirements

The AI should be able to remember:

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- Customer name
- Previous questions
- Previous answers
- Previous conversation
- Current conversation

Use PostgreSQL/Supabase as the persistent database.

Test

First ask:

"My name is Ahmed and I need help with my order."

Later ask:

"What is my name?"

The AI should retrieve the information from the database rather than relying only on temporary conversation memory.

Task 8 Natural Language → SQL AI Agent

Scenario

Build an AI agent that allows a business owner to ask questions about their database using normal English.

For example:

"How many leads did we receive this month?"

"Show me the top 5 leads."

"How many qualified leads came from Facebook?"

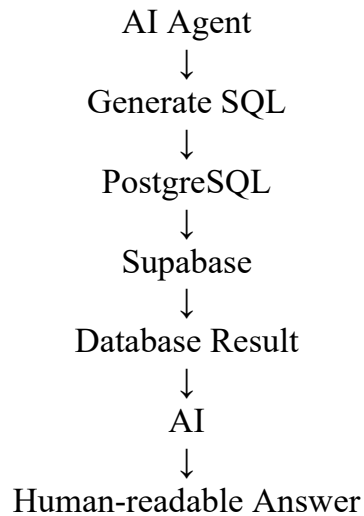
"What's our average lead score?"

Workflow

User Question



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Requirements

The agent should be capable of generating SQL for at least:

SELECT
WHERE
ORDER BY
COUNT
AVG
GROUP BY

Important

Students must **not allow the AI to blindly execute destructive queries.**

The AI should be restricted to safe/read-only queries for this task.

For example:

SELECT

is allowed.

But:

DROP TABLE
DELETE
TRUNCATE

should not be allowed.

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This task tests both **SQL knowledge** and **AI automation knowledge**.

Task 9 Automated Database Reporting

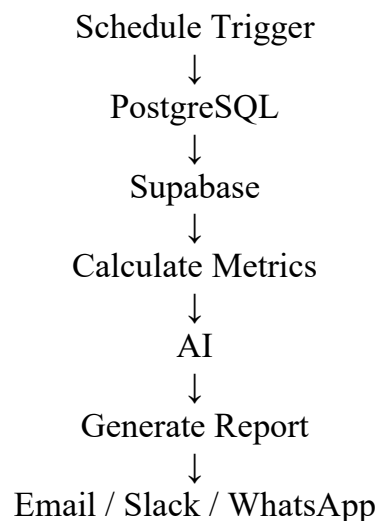
Scenario

A company wants an automated daily sales/lead report.

Every morning, n8n should:

1. Query PostgreSQL.
2. Retrieve data from Supabase.
3. Calculate statistics.
4. Send the data to an AI model.
5. Generate a business summary.
6. Send the report to the manager.

Workflow



The report should contain:

Total Leads
New Leads
Qualified Leads
Converted Leads
Average Lead Score
Top Lead Source
Top 5 Leads
Conversion Rate

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Example AI output

Daily Lead Report

Total Leads: 47

Qualified: 18

Converted: 6

Conversion Rate: 12.7%

Top Source:

LinkedIn

Observation:

LinkedIn generated fewer leads than the website,
but produced the highest percentage of qualified leads.

Final Submission Requirements

Students must submit:

SQL Tasks

- SQL queries
- Screenshots of execution
- Database/table screenshots

Paper Database Tasks

- Hand-drawn database structure

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- Tables
- Columns
- Primary Keys
- Foreign Keys
- Relationships
- ER-style diagram

n8n Tasks

For each workflow:

- n8n workflow screenshot
- PostgreSQL screenshot
- Supabase screenshot
- Successful execution screenshot
- SQL queries used
- Exported n8n workflow JSON, if required

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